SET NO. 23

EMPLOYEE PERFORMANCE ANALYSIS

Course:DSA0601 Data Handling and Visualization

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Question 1: Department Distribution Bar Chart

Aim:

To visualize the number of employees in each department using a bar chart.

Algorithm:

1. Enter employee department data.

2. Count employees in each department.

3. Draw a bar chart.

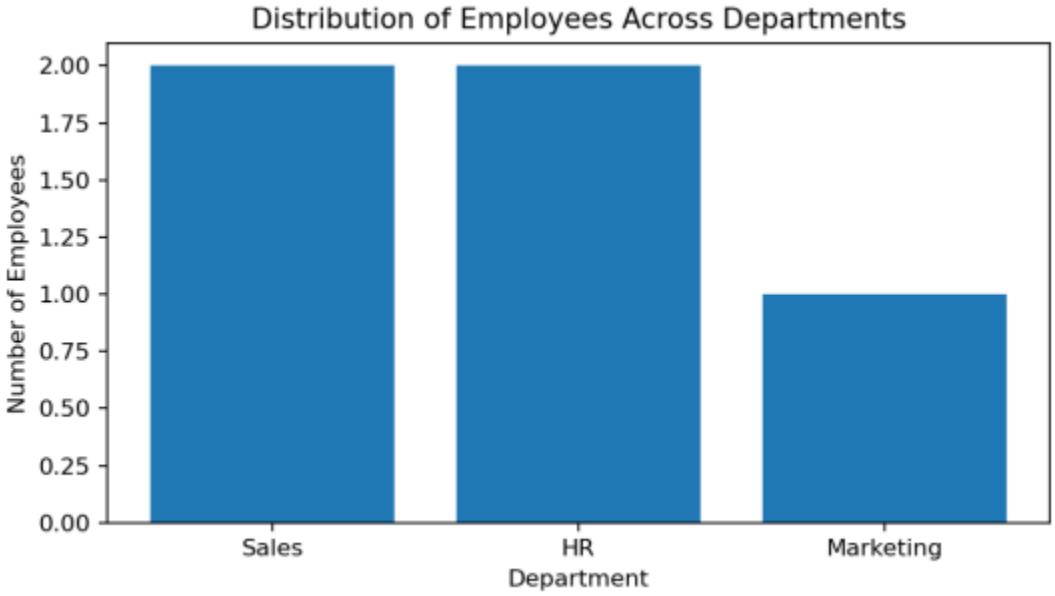
4. Label the x-axis, y-axis, and chart title.

Code (R):

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
count <- table(department)

barplot(count,
        main="Distribution of Employees Across Departments",
        xlab="Department",
        ylab="Number of Employees",
        col="skyblue")
```

Output: The bar chart shows Sales = 2 employees, HR = 2 employees, and Marketing = 1 employee.



Result: Sales and HR have the highest employee counts, while Marketing has one employee.

Question 2: Employee Performance Line Chart

Aim:

To display employee performance scores in a line chart.

Algorithm:

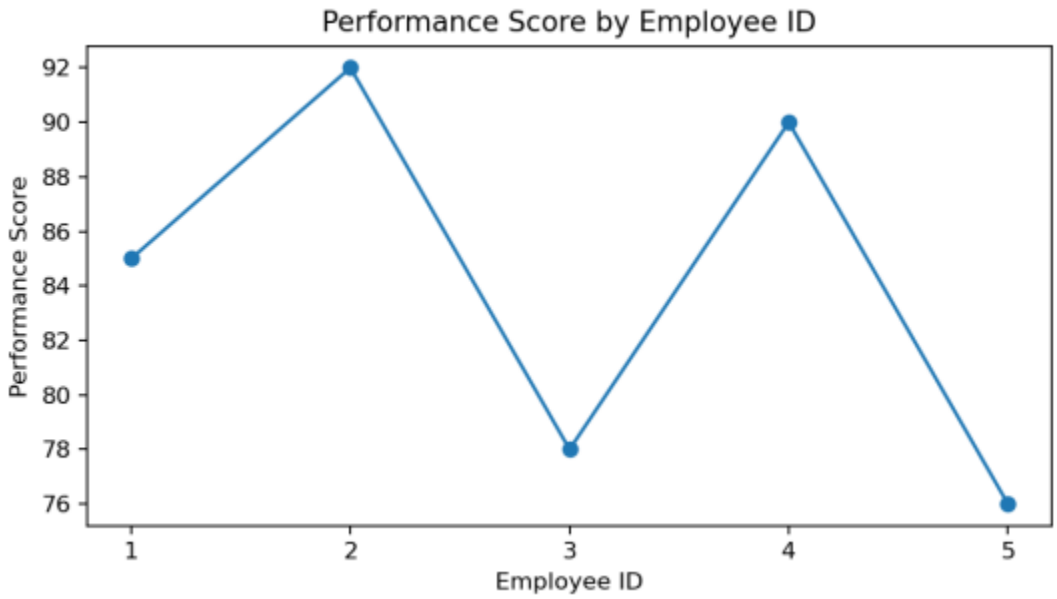
1. Store employee IDs and performance scores.
2. Plot the scores as a line.
3. Label both axes.
4. Add a title and display the chart.

Code (R):

```
employee_id <- c(1, 2, 3, 4, 5)
score <- c(85, 92, 78, 90, 76)

plot(employee_id, score, type="o", pch=19,
    main="Performance Score by Employee ID",
    xlab="Employee ID",
    ylab="Performance Score")
```

Output: A line chart displaying performance scores for employee IDs 1 to 5.



Result: Employee 2 has the highest score (92), and Employee 5 has the lowest score (76). A true performance trend over time requires dated observations.

Question 3: Tableau Dashboard

Aim: To build a dashboard combining department distribution and the relationship between years of service and performance score.

Algorithm:

1. Connect the employee dataset in Tableau.

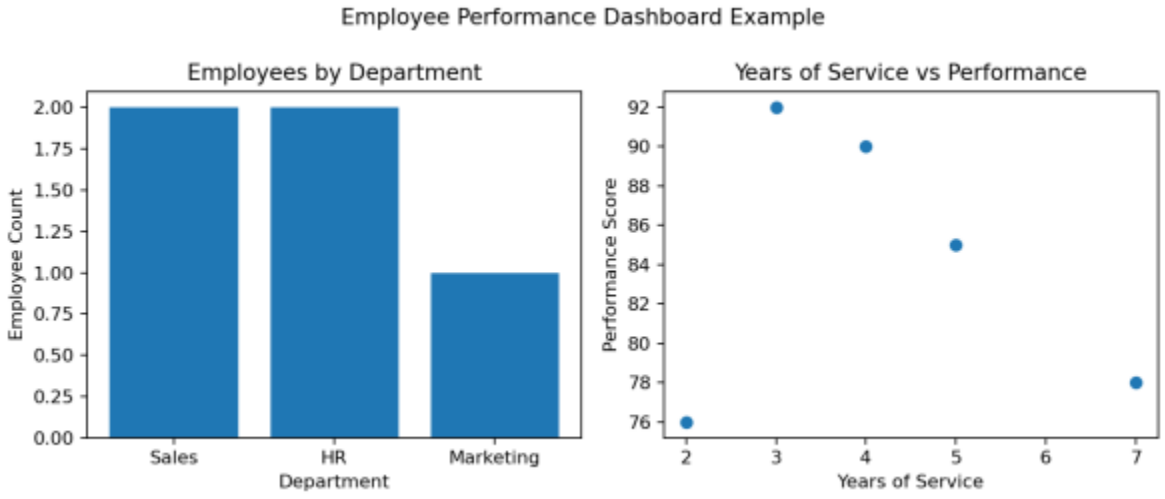
2. Create a bar chart with Department and employee count.

3. Create a scatter plot with Years of Service on Columns and Performance Score on Rows.

4. Add Employee ID to Detail.

5. Place both worksheets on a dashboard and add a title.

Output: A dashboard example with a department bar chart and a years-of-service versus performance scatter plot.



Result: The dashboard compares department sizes and helps inspect whether years of service are associated with performance scores.

Question 4: Employee Performance Data Table

Aim: To display employee performance data in a labelled table using R.

Algorithm:

1. Create vectors for employee ID, department, years of service, and performance score.

2. Combine them into a data frame.

3. Display the data frame with column names.

Code (R):

```
employee_id <- c(1, 2, 3, 4, 5)
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
years <- c(5, 3, 7, 4, 2)
score <- c(85, 92, 78, 90, 76)

employee_data <- data.frame(
  Employee_ID = employee_id,
  Department = department,
  Years_of_Service = years,
  Performance_Score = score
)
print(employee_data)
```

Output: A table containing employee ID, department, years of service, and performance score.

Employee Performance Data Table

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Result: The data frame presents all employee performance details in a clear, labelled table.